

Data foundations

What the job asks of you, how this programme runs, and how to read a table before you query it.

Tarek Atwan · Lead Instructor

Sun 20 Sep 2026

What we cover today

- 01 **Why this programme exists**
- 02 How the programme runs
- 03 Read the table before you query it
- 04 Set up BigQuery, GitHub and Power BI
- 05 Debrief and close



TAREK ATWAN

Data Science and Analytics Leader
5X Azure 3X AWS 1X Google

CERTIFICATIONS & ACCREDITATIONS

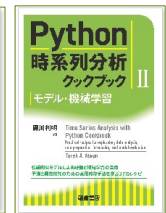
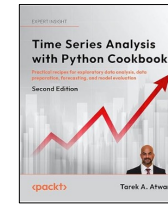
- **Azure** Fundamentals
- **Azure** Data Engineer
- **Azure** Data Scientist
- **Azure** AI Engineer
- **AWS** Cloud Practitioner
- **AWS** Machine Learning
- **Google** Digital Cloud Leader
- **Databricks** Developer Essentials
- **Databricks** Developer Foundations
- Astronomer **Apache Airflow** Fundamentals
- Confluent **Apache Kafka** Accreditation

Senior Technology Advisor & Corporate Trainer



SKILLS AND CORE AREAS OF FOCUS

- Executive Leadership
- Data Engineering
- Cloud ML and MLOps
- Big Data Analytics
- Machine Learning & Data Science
- Time Series Forecasting
- Deep Learning
- Data Visualization
- Python, R, and MATLAB
- Tensorflow & PyTorch
- SQL & NoSQL Databases
- Modern Data Architecture
- Data Governance
- Bilingual: Fluent in English & Arabic



CONSULTING:

An accomplished Analytics & Technology leader with over 20 years of experience in Data Science, Machine Learning, Data Analytics, Data Engineering, and Consulting Services, delivering strategic and cutting-edge data-driven solutions to clients across different functions and industries. Exceptional regional and international experience working with clients in North America, Europe, the Middle East & South Africa. A proven track record in building, training, and growing highly productive teams in data science, data analytics, and data engineering.



TEACHING:

7+ years of experience as a lead instructor & corporate trainer teaching Generative AI, AI Engineering, Machine Learning, Deep Learning, MLOps, Data Engineering, FinTech, Time Series Analysis & Forecasting, Exploratory Data Analysis, Data Visualization, AWS, Azure, Databricks, Snowflake, Spark, and SQL. The **University of Southern California**, **Arizona State University**, and **Pepperdine University**.



AUTHOR:

- Author of “*Time Series Analysis in Python Cookbook*” published by Packt Publishing in June 2022
- Book published in Japanese version through Asakura Publishing Company in December 2023
- Author of “*Time Series Analysis in Python Cookbook Second Edition*” to be published by Packt Publishing in January 2026

PART 01

Why this programme exists

The market you are training for, and the job you are training to do.

THE MARKET

170M

new jobs worldwide by 2030, against 92 million displaced.
Big data specialists lead the growth.

World Economic Forum, Future of Jobs Report 2025 (January 2025) · net growth of 78 million, 7% of total employment

What employers are asking for

01 **Analytical thinking is the top core skill**

Seven in ten employers call it essential, ahead of any tool. World Economic Forum, Future of Jobs Report 2025.

02 **Data roles grow faster than the labour market**

US projections to 2034: operations research analysts +21.5%, data scientists +33.5%, all occupations +3.1%.

03 **Postings asking for AI and data skills grew 69% in a year**

The job market as a whole grew 9%. PwC Global AI Jobs Barometer, June 2026, from over a billion job ads.

Jordan, and the market beyond it

Jordan

Jordan's ICT sector employs more than 46,000 people, and the sector association has set a target of about 100,000 jobs by 2033.

int@j, April 2025. Analyst work here sits inside banks, telecoms, retail chains, donor programmes and government digital projects.

The Gulf, and remote work

In the UAE, the share of job postings asking for AI skills rose from 1.0% in 2021 to 3.2% in 2025.

PwC UAE AI Jobs Barometer, June 2026. Analyst work is often done remotely, but this varies by the employer's policy.

WHAT THE SKILL IS WORTH

62%

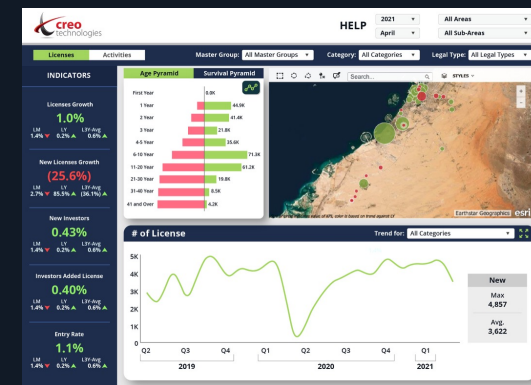
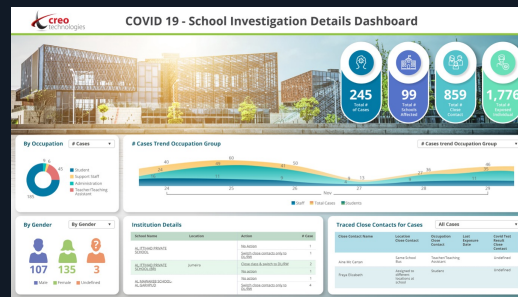


is the average wage premium for workers whose jobs ask for AI and data skills, up from 57%.

PwC 2026 Global AI Jobs Barometer (June 2026), from more than one billion job advertisements across 27 countries

BY THE END

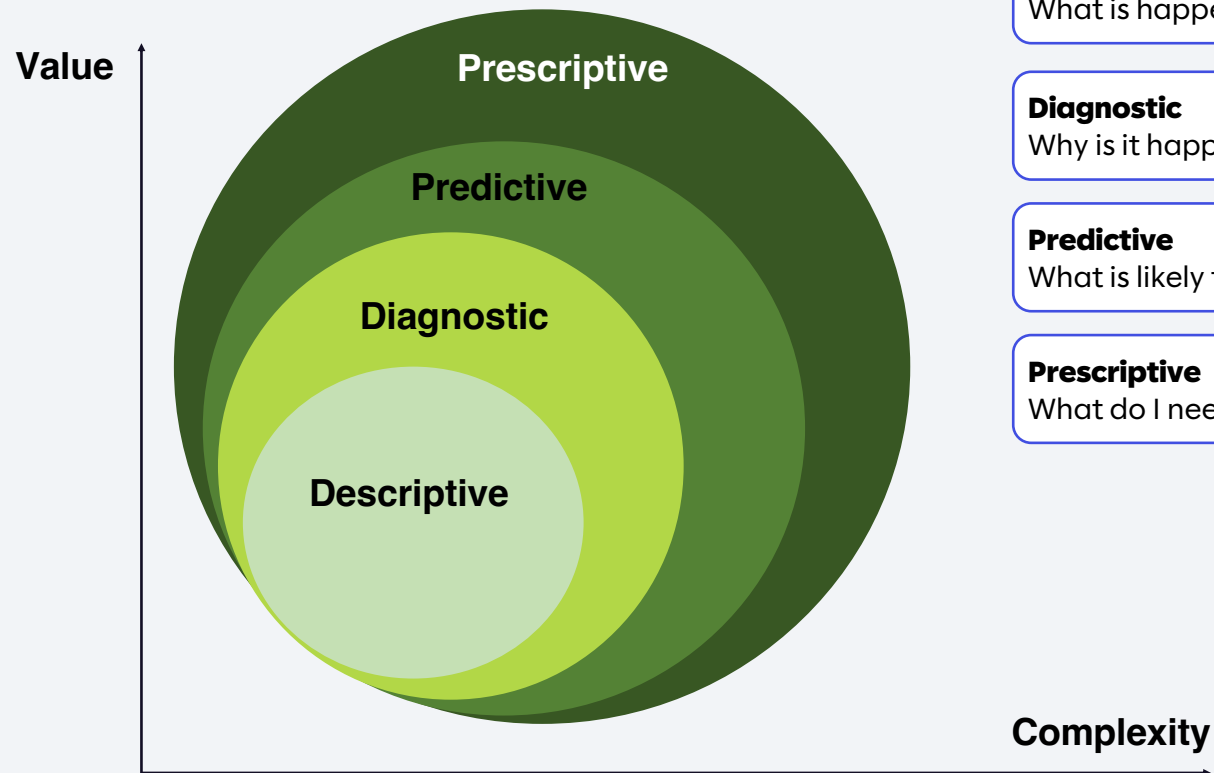
Work of this kind is the target



Three production reports. Each is a data model, a set of measures and a page layout.

SCOPE

Types of analytics



Descriptive

What is happening in my business?

Diagnostic

Why is it happening?

Predictive

What is likely to happen?

Prescriptive

What do I need to do?

Types of analytics

Descriptive

What happened? Sales by category last month, a KPI card, a report page someone opens every Monday. Most of modules 3 to 6.

Diagnostic

Why did it happen? Drill into the drop, compare segments, find where the number actually moved. Modules 4 and 5.

Predictive

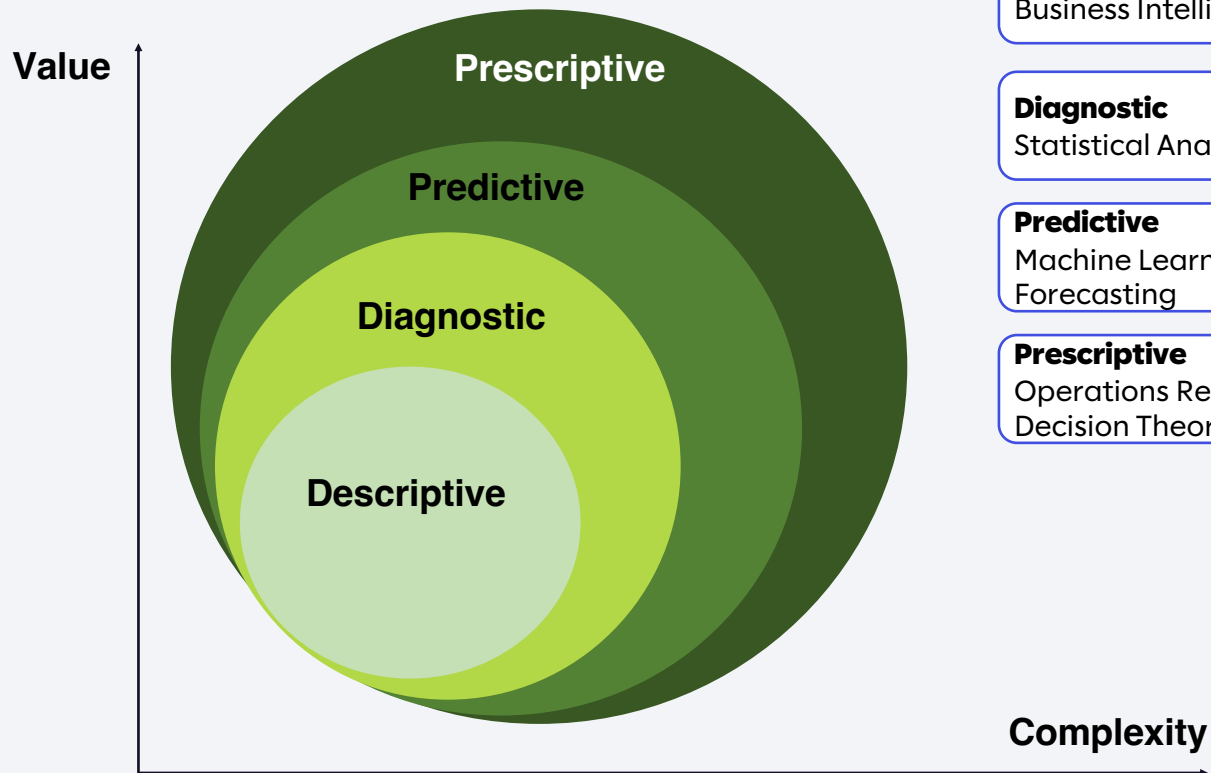
What is likely to happen? Which customers will churn. Module 7, twelve hours, built in BigQuery ML rather than Python.

Prescriptive

What should we do? Optimisation, simulation and decision models. Not taught here, but you will know where it sits.

SCOPE

Types of analytics



Descriptive

Business Intelligence, Dashboards, SQL Queries

Diagnostic

Statistical Analysis and Data Mining

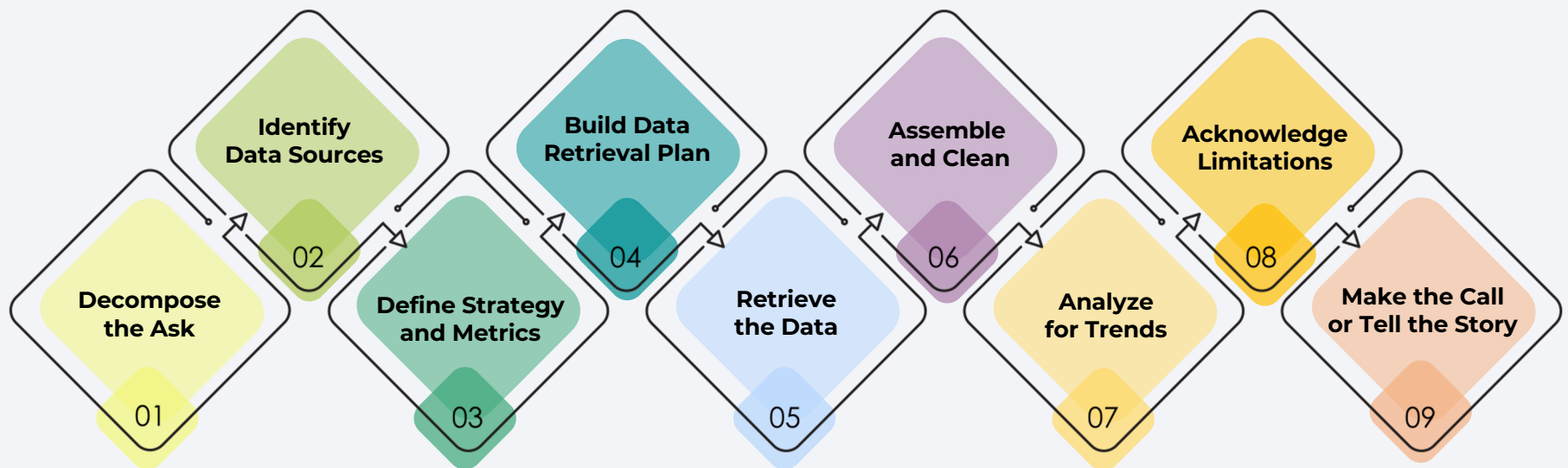
Predictive

Machine Learning, Deep Learning, Time Series
Forecasting

Prescriptive

Operations Research, Optimization, Simulations,
Decision Theory

The analytics paradigm



ROLES

Who does what on a data team

Role	What they own	Where you meet it here
Data engineer	Pipelines, storage, getting data to arrive reliably	Module 2, and BigQuery in module 7
Analytics engineer	Turning raw tables into trusted, documented ones	Module 3 Power Query, module 4 modelling
Data analyst	Answering business questions from data that exists	The whole programme. This is your target
BI developer	Models and reports other people open every day	Modules 4 to 6
Data scientist	Models that predict or classify	Module 7 using BigQuery ML

Titles vary by employer and the work overlaps. In a small company one person does all five.

PART 02

How this programme runs

Dates, modules, how you will learn, how you are graded, and the exam this prepares you for.

SCHEDULE

26 sessions, eight weeks

20 September to 12 November 2026. Every session is four hours with one break. 104 scheduled hours in total, of which 60 go to Power BI and exam preparation.

Green is instructor-led, split green and orange is a TA-led lab after the printed handoff, orange is TA-led throughout.

Schedule and Delivery Plan

ASAC Data Science & Analytics Programme — 20 September to 12 November 2026 — Power BI / PL-300 Track

26

Teaching days

4 hrs

Per day (2:00–6:00 PM)

104 hrs

Total scheduled hours

8 weeks

Programme duration

60 hrs

Power BI + exam prep

~200

Trainees

2:00–6:00 PM | Tarek + assigned TA groups | Break 3:35–4:00 PM | Split dates: Tarek teaches first; TAs lead the lab | PL-300 track

September 2026



Week 1: 2 days | Week 2: 3 days

TA 4:00 | Extended lab
Tarek 2:00–4:00, including the break → TAs 4:00–6:00
Practicals and assignment workshops

October 2026



Week 3: 3 days | Week 4: 2 days | Week 5: 3 days | Week 6: 3 days

TA 4:00 | Standard lab
Tarek 2:00–4:00, including the break → TAs 4:30–6:00
Demonstration, practice and group feedback

November 2026



Week 7: 5 days | Week 8: 5 days

TA 5:00 | Short practice
Tarek 2:00–5:00, including the break → TAs 5:00–6:00
Longer concept and demonstration sequence

■ Solid green: instructor-led D01–D02 ■ Half green / orange: Tarek + assigned TAs ■ Solid orange: TA-only D25–D26

Split colours show both delivery roles; the labelled handoff time controls the split. D25–D26 are TA-led throughout.

D01–D02 are instructor-led throughout. The first TA handoff is D03 on 27 September.

26 days × 4 hours = 104 scheduled hours, including 25-minute breaks. Delivery plan updated 19 September 2026.

Labs happen in class. Homework and assignments keep their published due dates. The D26 mock will follow the provider's confirmed timing and break rules.

MODULE MAP

Modules 1 to 4

01	Data foundations	8 hrs
02	SQL and GitHub	24 hrs
03	Power Query	16 hrs
04	Data modelling and DAX	16 hrs

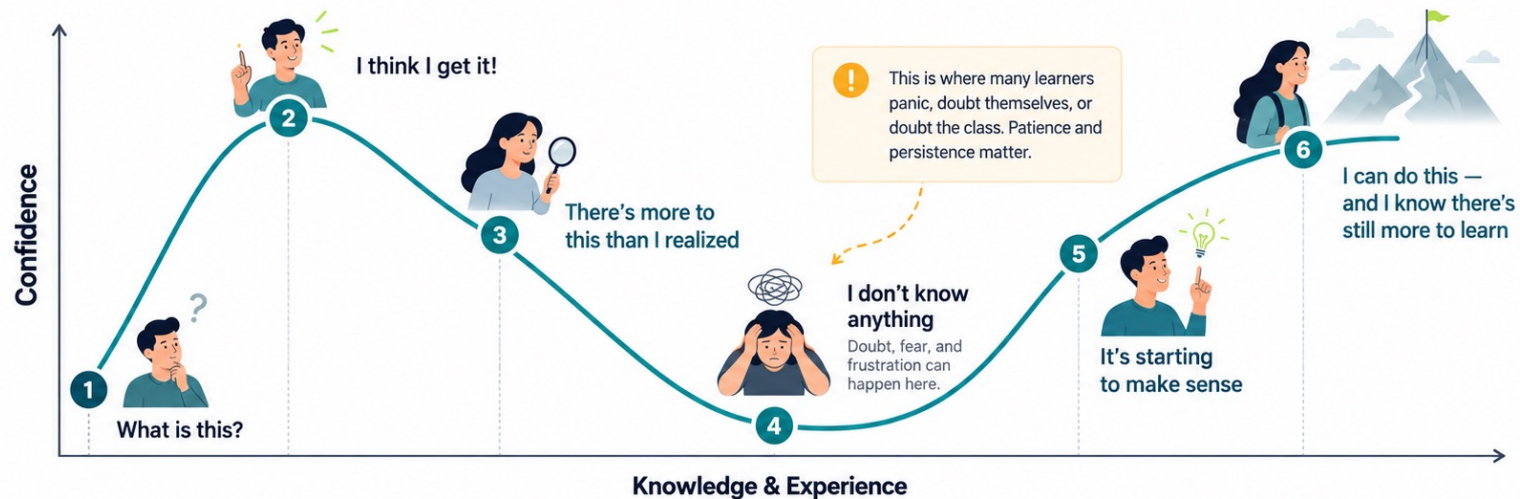
Modules 5 to 8

05	Visualisation and storytelling	16 hrs
06	Managing and securing Power BI	4 hrs
07	Machine learning and responsible AI	12 hrs
08	PL-300 review and career readiness	8 hrs

HOW YOU LEARN HERE

The Learning Curve Is Not Linear

Why learners often feel confident, then overwhelmed, then capable again



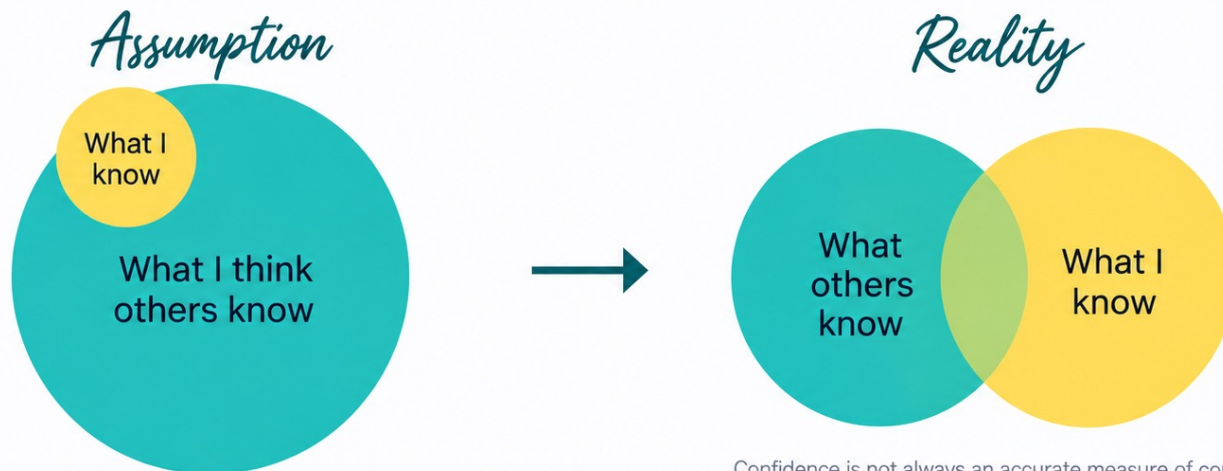
Great learning often feels uncomfortable in the middle.

With practice, support, and persistence, confusion turns into competence and humility.

— Confidence usually drops around week 3. Expect it, and ask for help early.

The Imposter Syndrome

What we assume others know versus what they actually know



Confidence is not always an accurate measure of competence.



Imposter syndrome distorts perception.

You often know more than you think, and others usually know less than you assume.

— You leave with an ASAC certificate, and an industry one too if you pass PL-300.

When a tool does your thinking

01 **Brain activity drops, and so does recall**

In an MIT Media Lab EEG study of 54 people, the group writing with an LLM showed the weakest brain connectivity and struggled to quote the essay they had just written. June 2025.

02 **The more you trust the tool, the less you check it**

A Microsoft Research and Carnegie Mellon survey of 319 knowledge workers: higher confidence in AI predicted less critical thinking, higher confidence in yourself predicted more. CHI 2025.

03 **The effect is strongest in your age group**

A 666-person UK study found heavier AI use tracked with lower critical thinking scores, with cognitive offloading as the mechanism, and 17 to 25 year olds most affected. Gerlich, Societies, 2025.

HOW YOU LEARN HERE

The words 19 sources keep using



HOW YOU LEARN HERE

Where AI fits in this programme

You will use AI tools in this field and so do I. The cost comes from reaching for one before you can read what it gives back. You cannot judge generated SQL until you can read SQL, and you cannot spot a wrong measure until you have written a right one.

In the coming weeks I will show you how I use these tools to learn faster, and the checks I run so that I keep the judgement.



In-class Labs vs Assignments (homework)

Lab

Done in class, with your TA in the room. It reinforces the concept you have just been shown.

Not graded, and it does not become homework. If you do not finish it, the reference solution is released at the debrief so you can complete it yourself.

Assignment or homework

Done on your own time, with a published brief, a due date and a rubric.

Graded. Four assignments and two in-session practicals carry 90% of the course grade. Professional practice carries the remaining 10%.

ASSESSMENT

How you are graded

01 **P1, SQL practical · 10%**

In session on D08, 7 October. Open book.

02 **A1 SQL portfolio 15%, A2 star schema 20%**

Due before the start of D10 and D14.

03 **P2 and A3 together 20%, A4 report and story 25%**

P2 in session on D16; A3 before the start of D17; A4 by the end of D21.

04 **Professional practice · 10%**

Continuous. Reviewed on D20; you write your charter in class on D24.

Who to ask, and when

01 **Your TA group is the first route**

You are assigned to one TA for the whole programme, and they are in the room for every session.

02 **Ask in the group channel before you ask privately**

If you are stuck on an install, others are too, and an answer in the channel is reusable.

03 **Escalate an access problem the same day**

An account fixed tonight costs you nothing. The same account fixed next week costs you a session.

What the exam measures

01 **Prepare the data · 25–30%**

Connect to sources, profile, clean and shape. This is Power Query, module 3.

02 **Model the data · 25–30%**

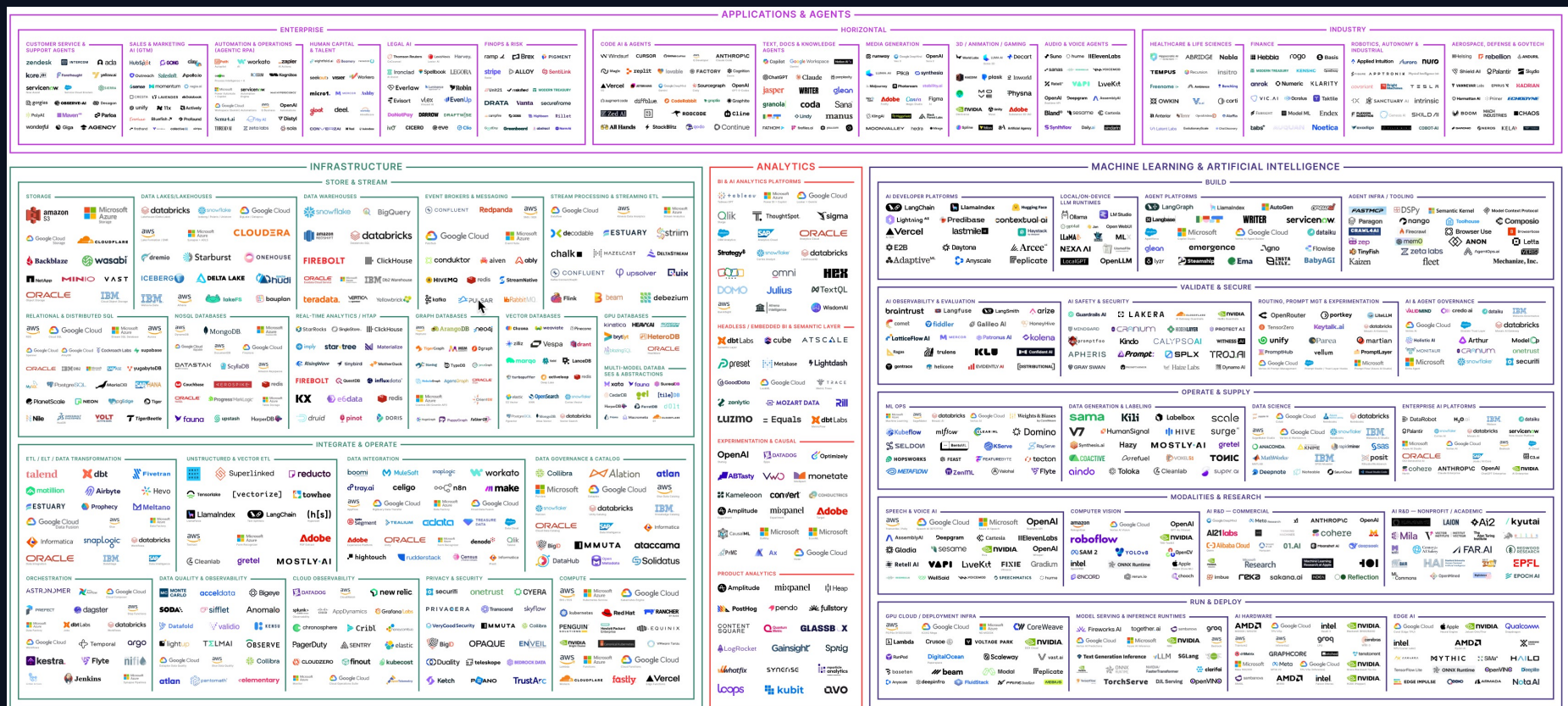
Star schemas, relationships, DAX measures and model performance. Module 4.

03 **Visualise and analyse the data · 25–30%**

Report design, interaction, accessibility and finding the story. Module 5.

04 **Manage and secure Power BI · 15–20%**

Workspaces, publishing, refresh and row-level security. Module 6.



You will not learn this. You will learn four of these well enough to be paid for it.

TOOLS

The tools you will use

01 **BigQuery · SQL and BigQuery ML**

Free, no credit card, runs in a browser. You query cloud tables from the third session.

02 **GitHub · your portfolio**

Where your SQL work is versioned, and the link you send an employer.

03 **Power BI Desktop · the core tool**

Power Query, the model, DAX and the report all live here. It needs Windows.

04 **Power BI Service · selected demonstrations**

Publishing and row-level security, as far as institutional access allows.

WHY SQL GETS 24 HOURS

50%



of data analyst postings in a 1,355-posting sample asked for SQL, ahead of Excel at 41% and Power BI at 25%.

365 Data Science, 1,355 Glassdoor US postings, April 2026 · a US sample: use the ranking, not the numbers

Break

 Twenty-five minutes

Post questions in the channel and I will pick them up when we return.

PART 03

Read the table before you query it

Grain, keys, dimensions, measures, and the problems you can see without any tool.

THINK
FIRST

5 minutes · on your own

Where is the best falafel in Amman?

TASK

Name one place, and the thing you would actually measure. Not “quality”. Something you could write in a column.

SOURCE

Where would the numbers come from? Something that already exists, reachable this afternoon, for free.

TEST

How would you know you were wrong? What would you expect to see that you are not seeing?

STEP 01

Decompose the ask

The same question, one word at a time.

STEP 01

Where is the **best** falafel
in Amman?

STEP 01

What does best mean?

01 Best tasting, or best value?

A sandwich at one dinar and a plate at six are not competing for the same thing.

02 Best at eight in the morning, or best at two in the morning?

The breakfast crowd and the after-midnight crowd rarely rate the same places.

03 Best according to whom?

You, your neighbourhood, everyone in Amman, or everyone who writes reviews online.



Best is an opinion. Before you can count it, you have to replace it with something you could put in a column.

STEP 01

**Where is the best falafel
in Amman?**

STEP 01

What counts as falafel?

01 The sandwich, or the plate?

Most places sell both, and people rate them as one experience rather than separately.



02 A falafel shop, or anywhere that sells it?

A restaurant with falafel on a long menu is not the same business as a falafel counter.



03 Made this morning, or reheated?

It matters, and no published source records it. Note it as a limitation and move on.

STEP 01

**Where is the best falafel
in Amman?**

STEP 01

Whose Amman?

01 The governorate, or the city?

They are different areas with different numbers of places, so they give different answers.

02 People who live here, or people who post?

Published reviews count only the people who chose to write one.

03 Open now, or open ever?

Places close. A list built last year answers a question about last year.

STEP 02

Identify data sources

What already exists, and what it will and will not tell you.

STEP 02

What is already published

Nobody needs to run a survey. Maps and delivery apps already hold a rating, a review count, a price level, a district and opening hours for most falafel places in Amman.

Each of those is a column you could put in a table, which is what makes it countable.

The screenshot shows a Google search for "best falafel in amman". The results page displays a list of restaurants, including "Abu Jbara - Madina Al-Monaw" and "Hashem Restaurant". A "talabat" logo is visible. Below the search results, a section titled "Top Falafel in Amman" lists 12 results. The first two results are detailed:

Rank	Restaurant Name	Rating	Reviews	Price Level	Status	Location
1	Al-Quds Falafel	4.5	388 reviews	Fast Food, Middle Eastern	Closed now	Al Akhdar
2	Hashem Restaurant Down Town	4.4	345 reviews	Fast Food, Mediterranean	Open now	Al Akhdar

The table also includes a column for "Location" with entries like "Al Akhdar" and "Al Swaifyeh".

STEP 03

The blueprint

The word	What we will actually count
Best	Average rating and review count, reported separately. Never averaged into one score.
Falafel	Places whose listed category is falafel, with at least thirty reviews.
Amman	Places inside Amman, grouped by district, so we can see whether the answer holds everywhere.

Nothing here is the real answer yet. It is the promise of what the answer will be made of.

STEPS 04 TO 06

Where the next steps are taught

01 **Retrieve the data**

Writing the query that pulls it. Module 2, you will learn how to write SQL.

02 **Assemble and clean**

Joining sources and fixing what is wrong. Module 3, Power Query.

03 **The part that starts today**

Knowing what one row of the result represents, before you trust any total built from it.

WHY THIS FIRST

\$12.9M

is what Gartner estimated poor data quality costs an average organisation each year.

Gartner, research published 2020 · treat this as an order of magnitude, not a precise current figure

THE QUESTION

Before you can answer this

Which product category produced the most completed-order revenue last month? You cannot answer it until you know what one row of the table in front of you represents.

If one row is an order item and you count rows as orders, your total is wrong and nothing warns you.

The winner changes with what you count

■ Every order item
■ Completed orders only

Shoes wins on every order item, because a cancelled order still has line items.



SAMPLE DATA

w01_retail · 5 orders, 12 items

Two tables, one order

orders

order_id	customer_id	order_date	status
1001	C-204	2026-08-03	completed
1002	C-118	2026-08-03	cancelled
1003	C-204	2026-08-11	completed

order_items

item_id	order_id	product_id	qty	price
9001	1001	P-77	2	19.50
9002	1001	P-14	1	8.00
9003	1002	P-77	3	19.50

Order 1001 has two rows in order_items. Count rows there and you are counting products sold, not orders taken.

Grain, key, dimension, measure

01 Grain

What one row represents. “One row per order item” is a grain. “Sales data” is not.

02 Candidate key

A column, or set of columns, that is unique and never blank across every row.

03 Dimension

A field you group or filter by: category, customer, order date, region.

04 Measure

A field you add up or average: quantity, price, revenue, days to ship.

The same data, counted two ways

Count rows in orders

Five rows, so five orders. Revenue for an order is the total of the items attached to it.

Correct when the question is about orders: how many did we take, what was the average order value, how many were cancelled.

Count rows in order_items

Twelve rows. Each row is one product on one order, so order 1001 appears twice.

Correct when the question is about products: which category sold most, how many units moved, which product is returned most often.

Problems you can see without a tool

01 The same customer appears twice

Two rows, two identifiers, one person. Every count of customers is now too high.

02 A required field is blank

A missing order date drops that row out of every date filter, and the report still looks fine.

03 One category is spelled three ways

“Shoes”, “shoes” and “SHOES ” group as three categories, and the real total is split across them.

Table-reading worksheet

ASAC Data Science & Analytics · D01 · 20 September 2026

For every table, fill in the five yellow cells: grain, candidate key, one dimension, one measure, one quality problem. Say why a key is plausible.

Name

TA group

TABLE 1

orders

order_id	customer_id	order_date	status
1001	C-204	2026-08-03	completed
1002	C-118	2026-08-03	cancelled
1003	C-204	2026-08-11	completed
1004	C-401	2026-08-18	completed
1005	C-118	2026-08-24	completed

Grain

What does one row represent?

Candidate key

Which column, or columns, is unique and never blank? Say why.

Dimension

Name one field you could group or filter by.

Measure

Name one field you could add up or average.

Quality problem

Name one thing that looks wrong, and how it could change an analysis.

For every table: grain, candidate key, one dimension, one measure, one quality problem.

PART 04

Get your tools working

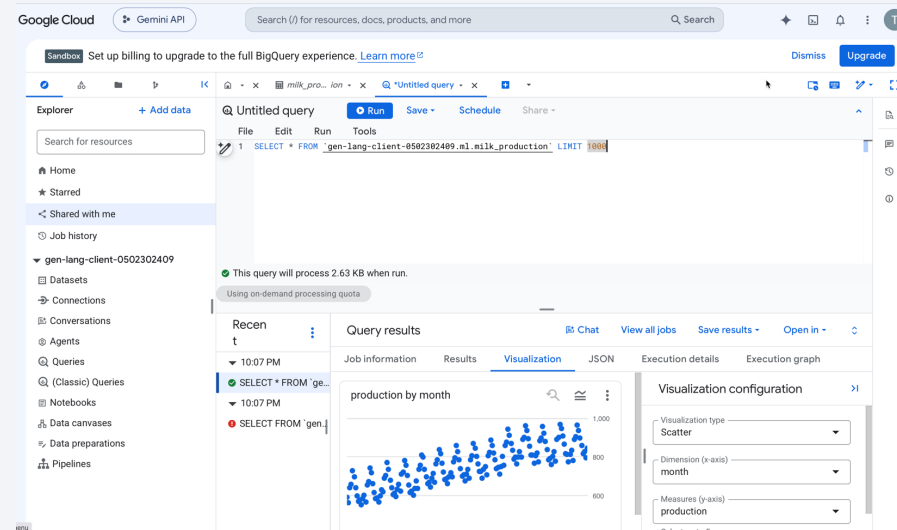
BigQuery, GitHub and Power BI Desktop, and a way to record what is blocked.

Check each tool once

01 Sign in to BigQuery Sandbox, open a public dataset and preview one table.

02 Sign in to GitHub and confirm you can create a repository.

03 Open Power BI Desktop (requires Windows OS)



YOUR TURN

at your own machine

D01 setup and table-reading worksheet

TASK

Complete the three tool checks, then annotate the three worksheet tables: grain, candidate key, one dimension, one measure, one quality problem.

FILES

W01-setup-checklist and W01-table-worksheet, both in the module 01 folder.

DONE WHEN

Every tool row says working or records a blocker with an owner, and all three tables are annotated.

Blocked on an account? Record it and keep going with the supplied file. Never share a password with anyone, including staff.

IF YOU ARE BLOCKED

When a tool will not open

01 **Record it, do not hide it**

Write the tool, the exact error text and the time on your checklist. That is the evidence we act on.

02 **Use the supplied file route**

Every check has a screenshot or file version, so a blocked account does not cost you the concept work.

03 **Hand your checklist in before you leave**

This session is the last scheduled time for setup. From D03 the class assumes working access.

RECAP

What you can do now

- Say what one row of a table represents
- Tell a dimension from a measure
- Name a plausible candidate key
- Spot a duplicate, a blank or an odd label

HOMEWORK

Week 1 set: get BigQuery, GitHub and Power BI Desktop working. Due before D03, 27 Sep. About 90 minutes.

Questions?

 Next session: Mon 21 Sep

D02: where data lives. Operational against analytical systems, and your first BigQuery query.